Table 1. Performance stratified by X-ray crystallographic resolution on the AsEP epitope_ratio test set (per-complex mean ± std). Complexes are binned into high (< 2.0 Å, 17 complexes), medium (2.0–3.0 Å, 75), and low (> 3.0 Å, 78) resolution. EpiFormer, MIPE, and WALLE are compared. N = number of complexes with available resolution data per method. EpiFormer maintains strong AUC (0.898–0.916) and F1 (0.464–0.622) across all bins with no catastrophic degradation at low resolution. MIPE has very few test samples with resolution data (3–6 per bin), limiting conclusions.

Resolution Bin	N	Method	AUC	AUPRC	F1	MCC	Prec.	Rec.
< 2.0 Å (high)	17	**EpiFormer**	**.908±.071**	**.691±.234**	**.622±.238**	**.553±.265**	**.555±.216**	.721±.269
	3	MIPE	.515±.212	.323±.168	.119±.205	.011±.203	.133±.231	.107±.185
	17	WALLE	.734±.204	.464±.325	.372±.146	.207±.209	.239±.113	**.964±.090**
2.0–3.0 Å (med)	75	**EpiFormer**	**.898±.080**	**.539±.253**	**.484±.235**	**.434±.243**	**.456±.224**	.553±.291
	6	MIPE	.674±.221	.251±.368	.163±.398	.145±.406	.159±.389	.167±.408
	75	WALLE	.719±.170	.300±.263	.238±.109	.137±.119	.141±.075	**.915±.146**
> 3.0 Å (low)	78	**EpiFormer**	**.916±.070**	**.508±.224**	**.464±.206**	**.437±.211**	**.433±.213**	.565±.271
	3	MIPE	.803±.128	.144±.072	.000±.000	.010 ± .011	.000±.000	.000±.000
	78	WALLE	.748±.172	.256±.203	.202±.096	.162±.121	.116±.062	**.932±.167**

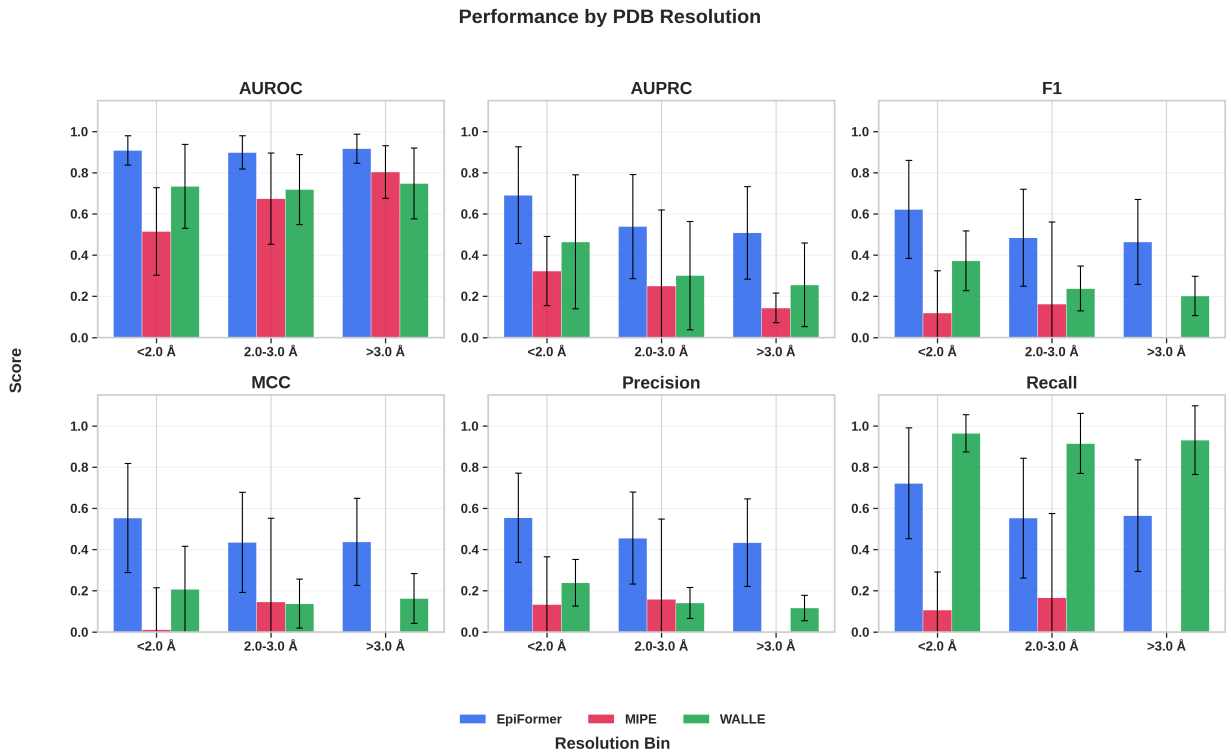

Figure 1. Per-complex AUC and F1 box plot distributions stratified by X-ray resolution (< 2.0 Å, 2.0–3.0 Å, > 3.0 Å) for EpiFormer, MIPE, and WALLE on the AsEP epitope_ratio test set. Each point represents one complex. EpiFormer maintains consistent median performance and tight interquartile range across all resolution bins, with no catastrophic degradation at low resolution (> 3.0 Å).